

# SAYAM SETHI

Email  | Google Scholar  | GitHub  | LinkedIn 

## EDUCATION

### PhD in Electrical and Computer Engineering

August 2023 – 2028 (tentative)

Advisor: Dr. Jonathan Baker; The University of Texas at Austin

Austin, Texas

GPA: 3.90/4.00

### Bachelor of Technology | Dept. of Computer Science and Engineering

July 2019 – May 2023

Indian Institute of Technology, Delhi

Delhi, India

GPA: 9.55/10.00

## PUBLICATIONS<sup>1</sup>

### Staying Sober: A Shot-Free Predictor of Logical Error Rates for Concatenated Quantum Error Correcting Codes

Oct 2026

*S. Sethi, M. Poster, A. Awasthi, J. Vízslai, and J.M. Baker*

*arXiv*

### Exploring Asymmetric QEC Code Concatenation

June 2026

*S. Sethi\*, M. Poster\*, A. Awasthi, W. Yang, J. Vízslai, and J.M. Baker*

*QEC '26 (poster)*

### Optimizing Parallel Execution of Commuting Pauli Product Rotations

May 2026

*S. Sethi, D. Nambisan, and J.M. Baker*

*arXiv*

### Price and Payoff: Non-Determinism in Fault Tolerant Quantum Computation

May 2026

*A. Awasthi, S. Sethi, S. Khan, G.S. Ravi, and J.M. Baker*

*QCE '26*

### INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures

April 2026

*S. Sethi, S. Khan, A. Awasthi, A. Anand, and J.M. Baker*

*QCE '26*

### Architecting Early Fault Tolerant Neutral Atoms Systems with Quantum Advantage

April 2026

*S. Khan, S. Sethi, K. Sahay, Y. Lin, J. Alnas, S. Kurapati, A. Anand, J.M. Baker, and K.R. Brown*

*Under review*

### Logical Compilation for Multi-Qubit Iceberg Patches

April 2026

*C. Mazzetti, S. Sethi, R. Rines, P. Gokhale, and J.M. Baker*

*Under review*

### Optimizing Logical Mappings for Quantum Low-Density Parity Check Codes

March 2026

*S. Sethi\*, S. Khan\*, M. Poster, A. Anand, and J.M. Baker*

*Under review*

### RESCQ: Realtime Scheduling for Continuous Angle Quantum Error Correction Architectures

March 2025

*S. Sethi, and J.M. Baker*

*ASPLOS '25*

### Cyclic Qubit Mappings

September 2024

*M. Poster, S. Sethi, and J.M. Baker*

*QCE '24*

### 3D-TemPo: Optimizing 3-D DRAM Performance Under Temperature and Power Constraints

February 2024

*S. Pandey, S. Sethi, and P.R. Panda*

*IEEE TCAD*

<sup>1</sup> \* denotes equal contribution

## INTERNSHIPS

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### **Infleqtion**

*Graduate Quantum Software Intern*

May 2026 - August 2026; 7 September 2026 - January 2027

### **IBM Research, TJ Watson Research Center**

*Quantum Research Scientist Intern*

Summer 2024 and Summer 2025

### **Quadeye Securities LLP, Gurugram**

*Systems Intern*

Jun – Jul 2022

## INVITED TALKS AND POSTERS

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### **Exploring Asymmetric QEC Code Concatenation (poster)**

*8th International Quantum Error Correction conference*

June 2026

*Santa Barbara, CA*

### **INJEQT: Improved Magic-State Injection Protocol for Fault-Tolerant Quantum Extractor Architectures**

*IBM Circuit and Compilers Group*

May 2026

*Virtual*

### **RESCQ: Realtime Scheduling for Continuous Angle Quantum Error Correction Architectures**

*IBM Circuit and Compilers Group*

*UT Austin CompArch Seminar*

April, May 2025

*Yorktown Heights, NY*

*Austin, TX*

## PROFESSIONAL SERVICES

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- Artifact Evaluation PC for 59th IEEE/ACM International Symposium on Microarchitecture (MICRO 2026).
- Junior PC for Symposium on Parallelism in Algorithms and Architectures (SPAA 2025).

## ACADEMIC ACHIEVEMENTS

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- **Texas Quantum Institute Graduate Fellowship (2025-27):** Awarded a fellowship for investigating classical overheads in FTQC compilation
- **IIT Delhi Semester Merit Award:** Secured for being in the **top 7 percentile** in **Semester I** of **2019-20** and **2021-22**
- **ICPC India Regionals, Amritapuri '21:** Secured **India Rank 47** and **Institute Rank 1** out of **4393 teams**
- **Google HashCode '22:** Secured **India Rank 12**, **Institute Rank 1** and **Global Rank 240** out of **10177 teams**
- **Indian National Mathematical Olympiad:** Qualified for **INMO 2018** (precursor to IMO) on being shortlisted in the Telangana region in **RMO 2017**
- **Kishore Vaigyanik Protsahan Yojana (KVPY):** Offered the fellowship in **SX stream (2018-19)** and **SA stream (2017-18)** with **All India Ranks** of **34** and **271** respectively

## TEACHING EXPERIENCE AND EXTRA-CURRICULAR ACTIVITIES

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### **Teaching Assistant**

- Grad TA for ECE 412 - Software Design & Implementation in Fall '26, course for ECE sophomores
- Grad TA for ECE 382V - Quantum Error Correction Theory & Systems in Spring '26
- Grad TA for ECE 312 - Software Design & Implementation in Fall '24, Spring '25, course for ECE sophomores
- Grad TA for ECE 306 - Introduction to Computing in Fall '23, introductory course for ECE freshers
- Lead TA for COL216 (**received the outstanding TA award**) - Computer Architecture in Spring '23, mandatory course for CSE sophomores
- Undergraduate TA for COL215 - Digital Logic and System Design in Fall '22, mandatory course for CSE sophomores

### **Algorithms and Coding Club, IIT Delhi**

- **Overall Coordinator, 2022-23:** Elevated the club's recognition from a Chapter to a Technical Club; organized events in Tryst, Technical Fest at IITD
- **Coordinator, 2021-22:** Conducted the Summer of CP and took live lectures on YouTube; guided executives
- **Executive, 2020-21:** Authored various problems for Competitive Programming contests conducted by the club

## TECHNICAL SKILLS

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**Programming Languages:** C, C++, Python, Java, VHDL, Assembly, Standard ML, Octave, MATLAB, R  
**Libraries/Frameworks and Tools:** Qiskit, Stim, Sniper, HotSpot, CoMeT, Vivado, L<sup>A</sup>T<sub>E</sub>X, git